

Mutual Fund Analytics

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Table of Contents

Table of Contents	2
Executive Summary	4
1 Introduction and Project Objectives	5
1.1 Project Background	5
1.2 Project Objectives	5
2 Data Sources	6
2.1 Dataset Overview	6
2.2 Raw and Processed Datasets	6
2.3 Raw Data	6
2.4 Processed Data	7
2.5 API Data	7
3 ETL Design and Data Pipeline	8
3.1 ETL Overview	8
3.2 Project Architecture	8
3.3 Data Ingestion	8
3.4 Data Cleaning	8
4 Exploratory Data Analysis	9
4.1 NAV Trend Analysis	9
4.2 AUM Analysis	9
4.3 SIP Inflow Analysis	10
4.4 Category-wise Inflow Heatmap	10
4.5 Investor Demographics	11
4.6 Geographic Distribution	12
4.7 Folio Count Growth	12
4.8 Correlation Analysis	12
4.9 Sector Allocation	13
5 Fund Performance Analysis	14
5.1 Daily Returns	14
5.2 CAGR	14
5.3 Sharpe Ratio	14
5.4 Sortino Ratio	14

5.5	Alpha and Beta	14
5.6	Maximum Drawdown	14
6	Advanced Analytics and Risk Analysis	15
6.1	Value at Risk	15
6.2	Conditional Value at Risk	15
6.3	Rolling Sharpe Ratio	15
6.4	Investor Cohort Analysis	15
6.5	SIP Continuation Analysis	15
6.6	Fund Recommendation Model	15
6.7	Sector Concentration Risk	15
7	Power BI Dashboard	16
7.1	Dashboard Overview	16
7.2	Industry Overview	16
7.3	Fund Performance	16
7.4	Investor Analytics	17
7.5	SIP and Market Trends	18
7.6	Interactivity and Branding	18
8	Findings and Conclusions	19
8.1	Key Findings	19
8.2	Recommendations	19
8.3	Limitations	20
8.4	Conclusion	20

Executive Summary

This project presents an end-to-end data analytics solution for analysing the Indian mutual fund industry. The project covers the complete analytics lifecycle, including data ingestion, data cleaning, exploratory data analysis, fund performance evaluation, risk analysis, advanced analytics and interactive dashboard development.

The analysis uses mutual fund datasets containing NAV history, fund information, assets under management, SIP inflows, investor transactions, scheme performance, portfolio holdings, benchmark indices and other industry-level information. Python and Pandas were used for data ingestion, cleaning, exploration and performance analytics. SQLite was used for structured data storage and SQL-based analysis. Power BI was used to develop an interactive dashboard for business users.

The project also implements financial performance and risk metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR) and rolling Sharpe Ratio. The final solution provides insights into mutual fund performance, investor behaviour, industry growth, risk characteristics and portfolio concentration.

Chapter 1

Introduction and Project Objectives

1.1 Project Background

The mutual fund industry generates large volumes of financial and investor data. Analysing this data can help investors, analysts and financial organisations understand fund performance, risk, investor behaviour and market trends.

The objective of this project is to build a complete analytics pipeline for mutual fund data and present the results through an interactive business intelligence dashboard.

1.2 Project Objectives

The major objectives of the project are:

- Ingest mutual fund datasets using Python.
- Fetch live NAV data using a public mutual fund API.
- Clean and validate the available datasets.
- Store structured data in a SQLite database.
- Perform exploratory data analysis.
- Calculate fund performance and risk metrics.
- Compare fund performance against market benchmarks.
- Perform advanced risk and investor analytics.
- Build an interactive Power BI dashboard.
- Present recommendations and business insights.

Chapter 2

Data Sources

2.1 Dataset Overview

The Bluestock Mutual Fund Analytics project uses multiple datasets covering mutual fund schemes, historical NAV, assets under management, investor transactions, SIP activity, portfolio holdings and market benchmarks.

The primary data consists of 10 CSV datasets provided for the project. These datasets were initially stored in the `data/raw/` directory. During the ETL process, the datasets were cleaned, validated, standardised, and stored in the `data/processed/` directory for further analysis.

2.2 Raw and Processed Datasets

Raw Dataset	Processed Dataset	Purpose
01_fund_master.csv	clean_fund_master.csv	Fund and scheme master information
02_nav_history.csv	clean_nav.csv	Historical NAV values for mutual fund schemes
03_aum_by_fund_house.csv	clean_aum_by_fund_house.csv	Assets under management by fund house
04_monthly_sip_inflows.csv	clean_monthly_sip.csv	Monthly SIP industry inflows
05_category_inflows.csv	clean_category.csv	Category-wise mutual fund inflows
06_industry_folio_count.csv	clean_industry_folio_count.csv	Industry folio count and investor growth
07_scheme_performance.csv	clean_performance.csv	Scheme-level performance and risk metrics
08_investor_transactions.csv	clean_transactions.csv	Investor transaction and demographic information
09_portfolio_holdings.csv	clean_portfolio.csv	Fund portfolio and sector allocation
10_benchmark_indices.csv	clean_benchmark.csv	Market benchmark index values

Table 2.1: Raw and processed datasets used in the project

2.3 Raw Data

The original datasets were preserved in the `data/raw/` directory. Keeping the raw files unchanged provides a reference point for data validation and allows the ETL process to be repro-

duced when required.

The raw datasets contain information at different levels of granularity. For example, NAV history contains daily observations, while AUM, SIP inflow, and folio datasets contain periodic industry-level observations.

2.4 Processed Data

After data cleaning and validation, the processed datasets were stored in the `data/processed/` directory. The main transformations included date standardisation, duplicate removal, numeric validation, missing-value handling, transaction type standardisation, and validation of financial values.

The processed NAV dataset contains approximately 46,000 records covering 40 mutual fund schemes. The historical NAV data spans from January 2022 to May 2026. The processed investor transaction dataset contains more than 32,000 transaction records and was used for investor behaviour, demographic, and SIP continuity analysis. The processed scheme performance dataset contains information for 40 schemes and was used for fund ranking and performance analysis.

2.5 API Data

In addition to the provided CSV datasets, NAV data was collected from the public MFAPI REST API. The API was used to understand REST-based financial data ingestion and to obtain NAV information for selected mutual fund schemes.

The following schemes were used for API-based NAV collection:

Scheme	AMFI Code
HDFC Top 100	125497
SBI Bluechip	119551
ICICI Bluechip	120503
Nippon Large Cap	118632
Axis Bluechip	119092
Kotak Bluechip	120841

Table 2.2: Selected schemes used for API-based NAV ingestion

The API responses were parsed from JSON format using Python and converted into structured CSV files. These files were retained in the raw data area before further processing.

Chapter 3

ETL Design and Data Pipeline

3.1 ETL Overview

The project follows an Extract, Transform and Load (ETL) workflow for preparing mutual fund data for analysis and reporting.

1. **Extract:** Load datasets from CSV files and fetch selected NAV data using the MFAPI REST API.
2. **Transform:** Clean, validate, standardise and prepare the datasets for analysis.
3. **Load:** Store processed datasets as CSV files and load analytical tables into SQLite.
4. **Analyse:** Perform financial analytics using Python and SQL.
5. **Visualise:** Present key business insights through Power BI dashboards.

3.2 Project Architecture

The overall data flow is:

Raw Data → ETL / Cleaning → Processed Data → SQLite → Analytics → Power BI

3.3 Data Ingestion

Python and Pandas were used to ingest the provided CSV datasets and inspect their shapes, columns, data types, and sample records. The MFAPI REST API was additionally used to fetch NAV information for selected mutual fund schemes.

3.4 Data Cleaning

The main data-quality and transformation steps included:

- Standardising date formats, data types and removing duplicate records.
- Validating numeric values and handling missing values where required.
- Sorting records by fund and date for time-series analysis.

Chapter 4

Exploratory Data Analysis

4.1 NAV Trend Analysis

Historical NAV trends were analysed across the mutual fund schemes. The analysis was used to understand long-term fund movement, market recovery periods, rallies and corrections.

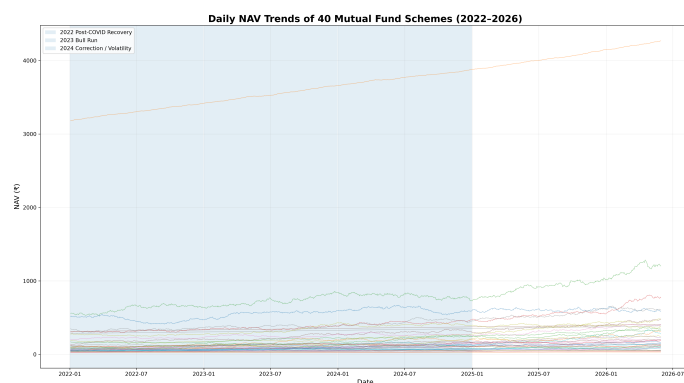


Figure 4.1: NAV Trend Analysis of Mutual Fund Schemes

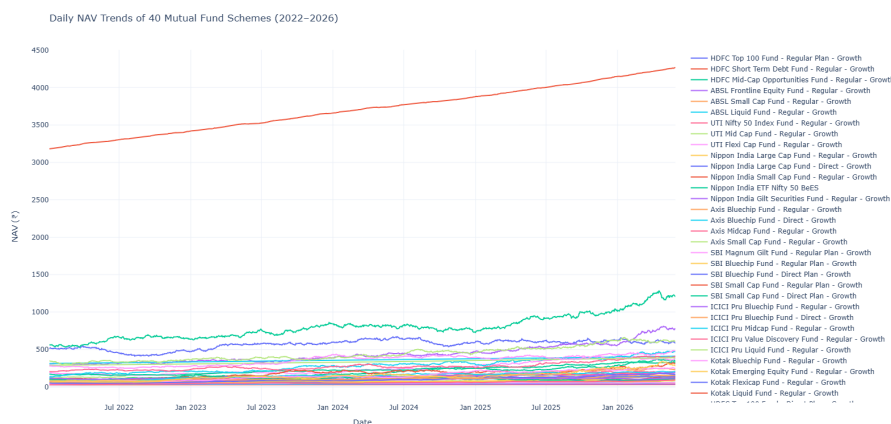


Figure 4.2: NAV Trend Analysis of Mutual Fund Schemes - Additional View

4.2 AUM Analysis

Assets under management were analysed by fund house and year. This helped identify major asset management companies and changes in industry-level assets.

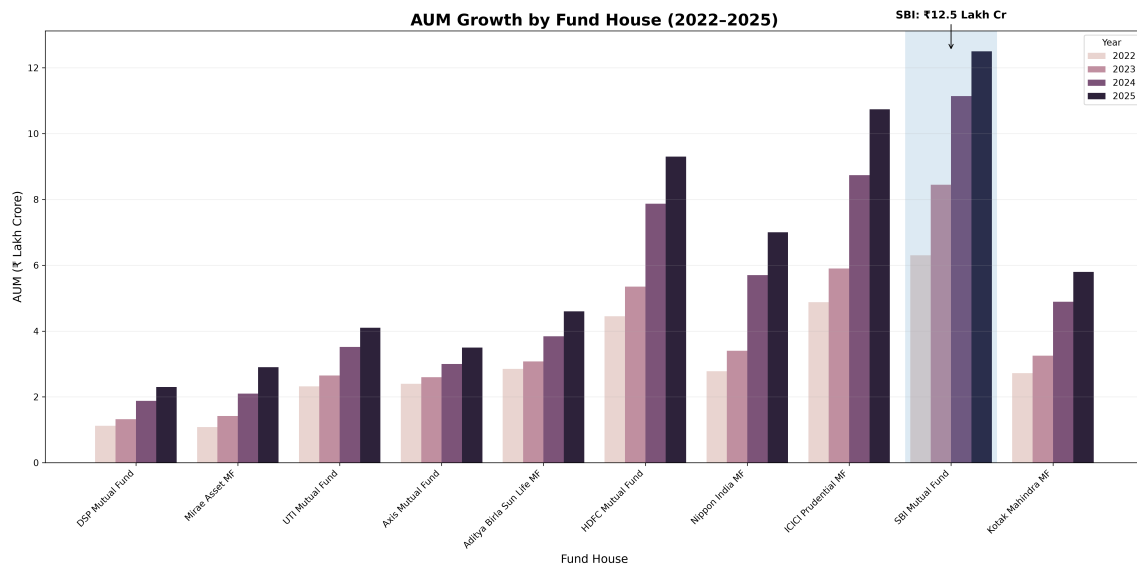


Figure 4.3: AUM Growth by Fund House

4.3 SIP Inflow Analysis

Monthly SIP inflows were analysed to identify long-term growth patterns and important industry milestones.

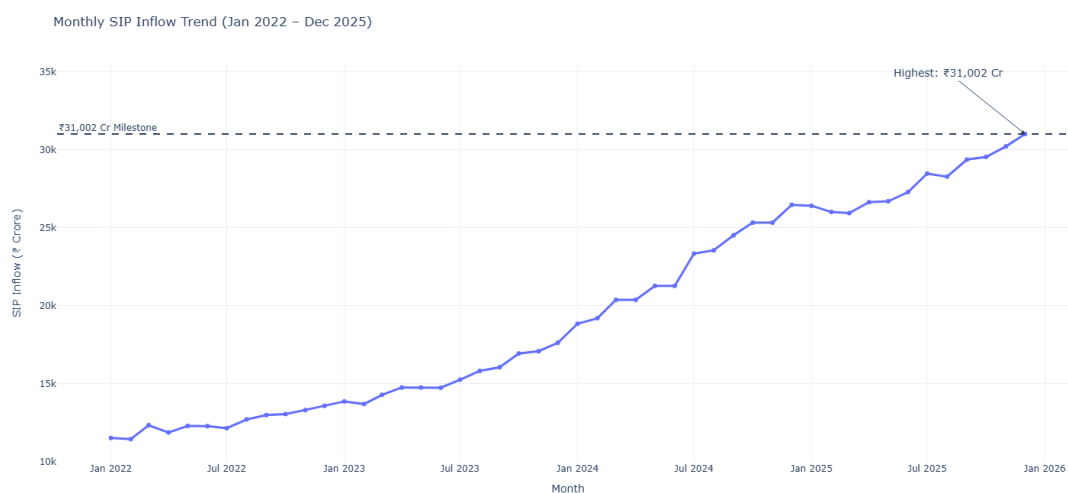


Figure 4.4: Monthly SIP Inflow Trend

4.4 Category-wise Inflow Heatmap

Category-wise mutual fund inflows were analysed across months to identify changes in investor preferences and category-level trends.

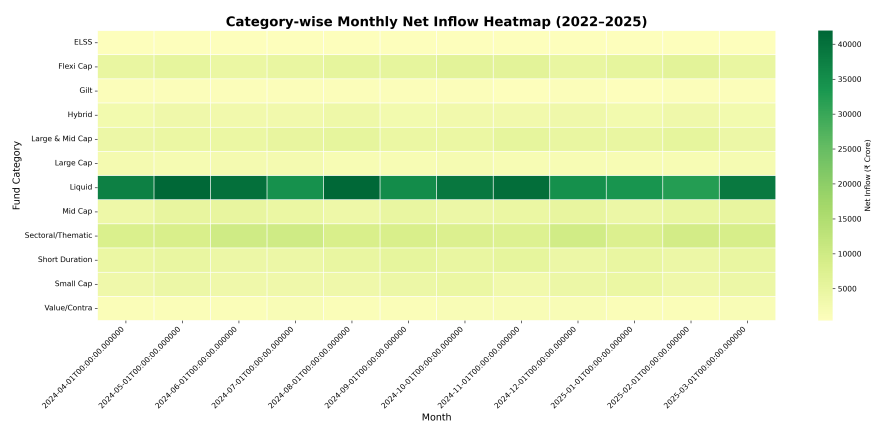


Figure 4.5: Category-wise Mutual Fund Inflow Heatmap

4.5 Investor Demographics

Investor transactions were analysed across age groups, transaction types and other demographic dimensions.

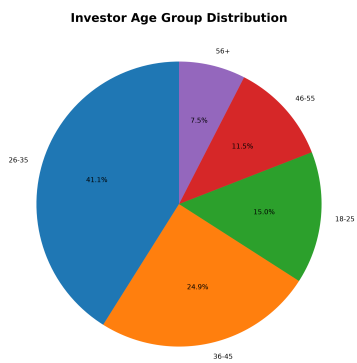


Figure 4.6: Investor Age Group Distribution

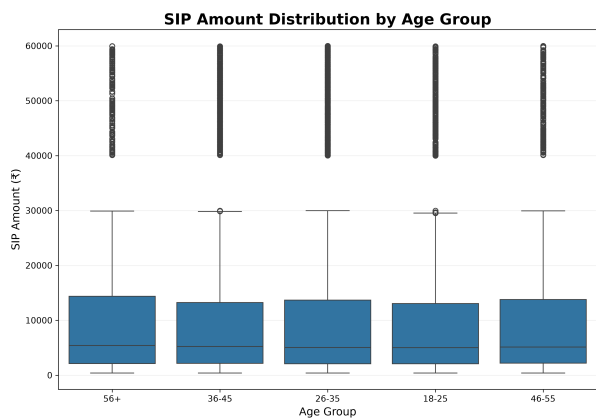


Figure 4.7: SIP Amount Distribution by Age Group

4.6 Geographic Distribution

The geographic distribution of mutual fund investments was analysed using SIP amounts by state and the distribution between T30 and B30 locations. This analysis helps understand the regional participation of investors.

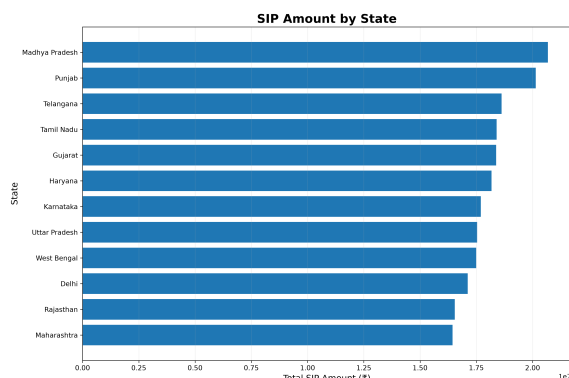


Figure 4.8: SIP Amount Distribution by State

SIP Amount Distribution: T30 vs B30 Cities

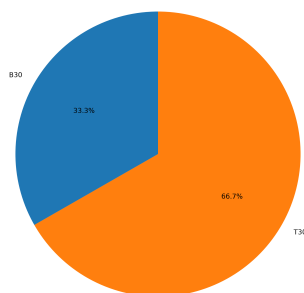


Figure 4.9: Geographic Distribution: T30 vs B30

4.7 Folio Count Growth

Industry folio counts were analysed to understand the growth in mutual fund participation over time.

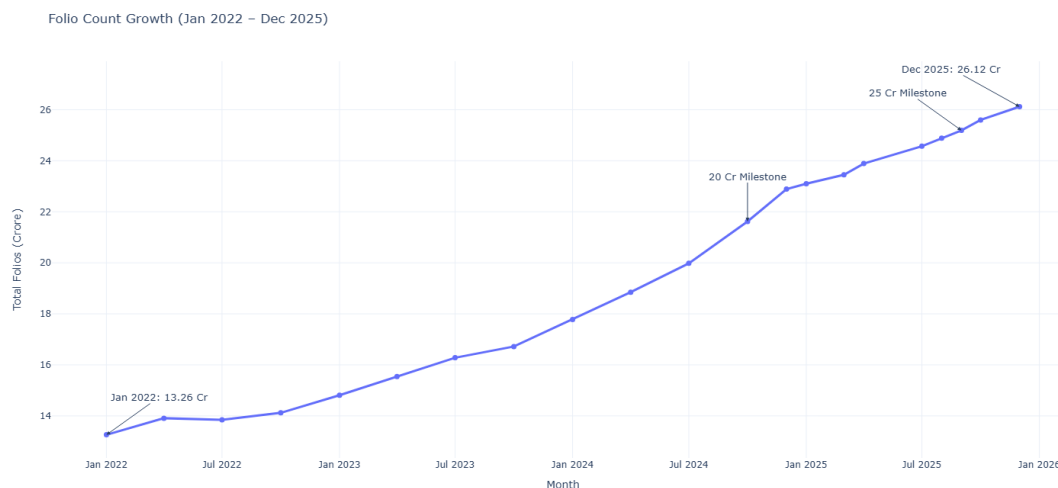


Figure 4.10: Mutual Fund Folio Count Growth

4.8 Correlation Analysis

Correlation between selected fund returns was calculated to understand the relationship between fund movements.

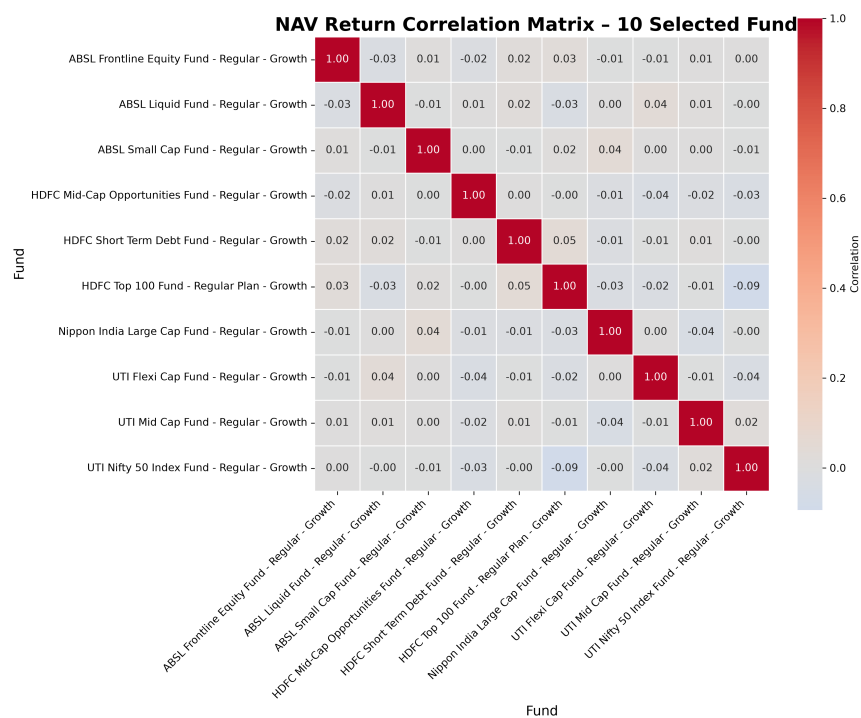


Figure 4.11: Correlation Matrix of Selected Fund Returns

4.9 Sector Allocation

Portfolio holdings were analysed to understand the distribution of investments across sectors.

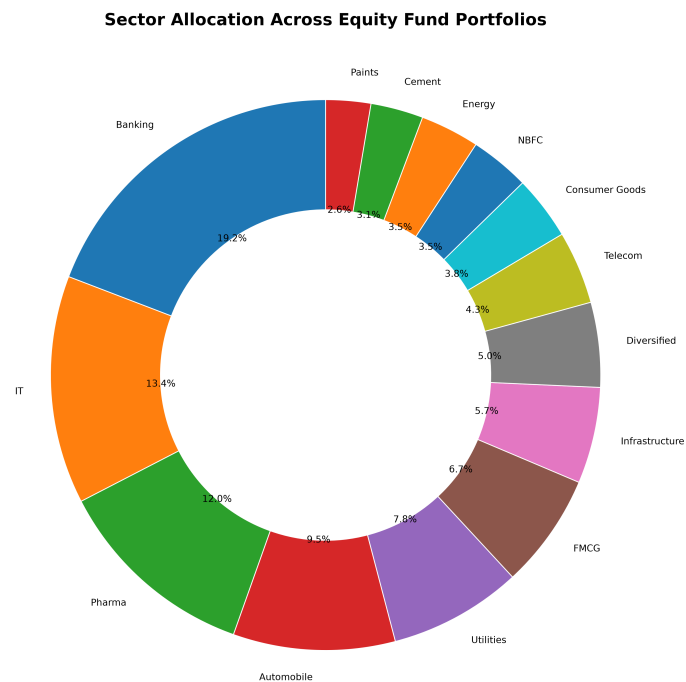


Figure 4.12: Sector Allocation Across Equity Fund Portfolios

Chapter 5

Fund Performance Analysis

5.1 Daily Returns

Daily fund returns were calculated using:

$$R_t = \frac{NAV_t}{NAV_{t-1}} - 1$$

5.2 CAGR

Compound Annual Growth Rate was calculated for different investment periods.

$$CAGR = \left(\frac{NAV_{end}}{NAV_{start}} \right)^{1/n} - 1$$

5.3 Sharpe Ratio

The Sharpe Ratio was calculated to measure risk-adjusted return.

$$Sharpe = \frac{R_p - R_f}{\sigma_p}$$

An annualised risk-free rate proxy of 6.5% was used in the analysis.

5.4 Sortino Ratio

The Sortino Ratio was calculated using downside volatility, focusing only on negative return observations.

5.5 Alpha and Beta

Fund returns were compared against benchmark returns using linear regression.

Alpha measures excess performance while Beta measures sensitivity to benchmark movements.

5.6 Maximum Drawdown

Maximum Drawdown measures the largest decline from a historical peak.

$$Drawdown_t = \frac{NAV_t}{RunningMax_t} - 1$$

Chapter 6

Advanced Analytics and Risk Analysis

6.1 Value at Risk

Historical Value at Risk at the 95 using the fifth percentile of daily returns.

6.2 Conditional Value at Risk

Conditional VaR was calculated as the average return of observations below the VaR threshold.

6.3 Rolling Sharpe Ratio

A rolling 90-day Sharpe Ratio was calculated for selected funds to observe changes in risk-adjusted performance over time.

6.4 Investor Cohort Analysis

Investors were grouped according to their first transaction year. The analysis compared investment behaviour between cohorts.

6.5 SIP Continuation Analysis

Investors with multiple SIP transactions were analysed based on the average gap between transactions. Investors with longer gaps were identified as potentially at-risk SIP investors.

6.6 Fund Recommendation Model

A simple recommendation model was developed using investor risk appetite and fund risk grades. The model recommends top funds based on Sharpe Ratio within the matching risk category.

6.7 Sector Concentration Risk

Sector concentration was measured using the Herfindahl-Hirschman Index (HHI). A higher HHI indicates greater concentration in fewer sectors, while a lower value indicates better diversification.

$$HHI = \sum_{i=1}^n w_i^2$$

Chapter 7

Power BI Dashboard

7.1 Dashboard Overview

The dashboard consists of four pages covering industry trends, fund performance, investor behaviour, and SIP and market trends. Interactive slicers, tooltips, and drill-through functionality allow users to explore the data at different levels of detail.

7.2 Industry Overview

An interactive Power BI dashboard was developed to present the major findings of the analysis. The first dashboard page presents KPIs including AUM, SIP inflows, folio count and scheme information. The page includes KPI cards showing Total AUM of approximately Rs.62.74L Cr, SIP inflows of Rs.31K Cr, 26.12 Cr folios, and 2K schemes. An industry AUM line chart and a top 10 fund house bar chart provide a view of asset growth and the contribution of AMCs.

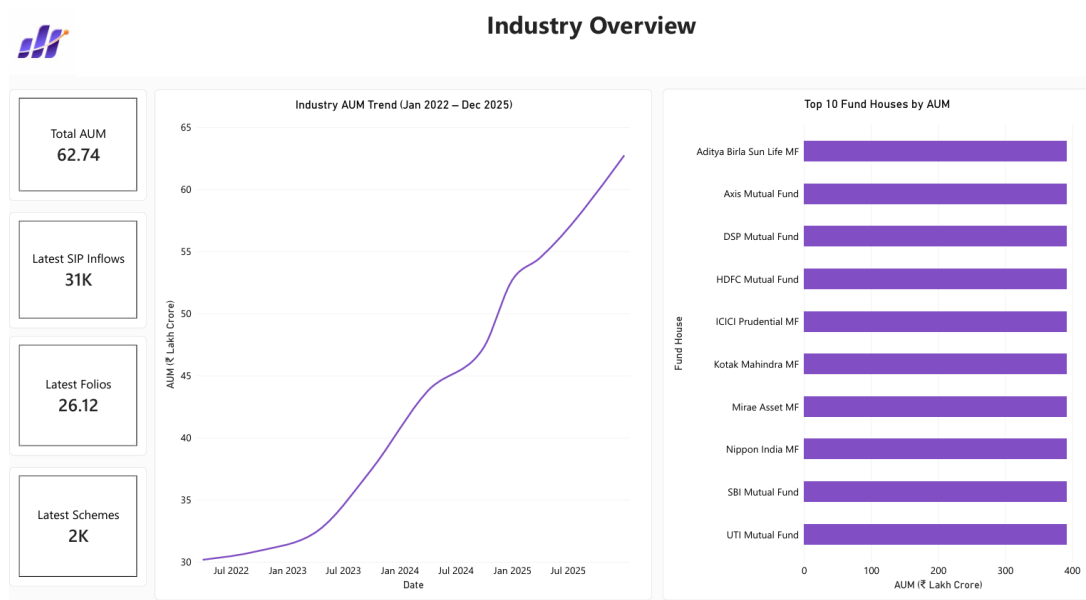


Figure 7.1: Power BI Dashboard – Industry Overview

7.3 Fund Performance

The second page presents fund-level performance using scatter plots, scorecards, NAV trends, benchmark comparisons, and interactive slicers. The scatter plot compares fund return with

risk, with AUM represented by bubble size. A sortable fund scorecard helps identify stronger performing schemes, while the NAV versus benchmark line chart allows users to compare the selected fund against its benchmark.

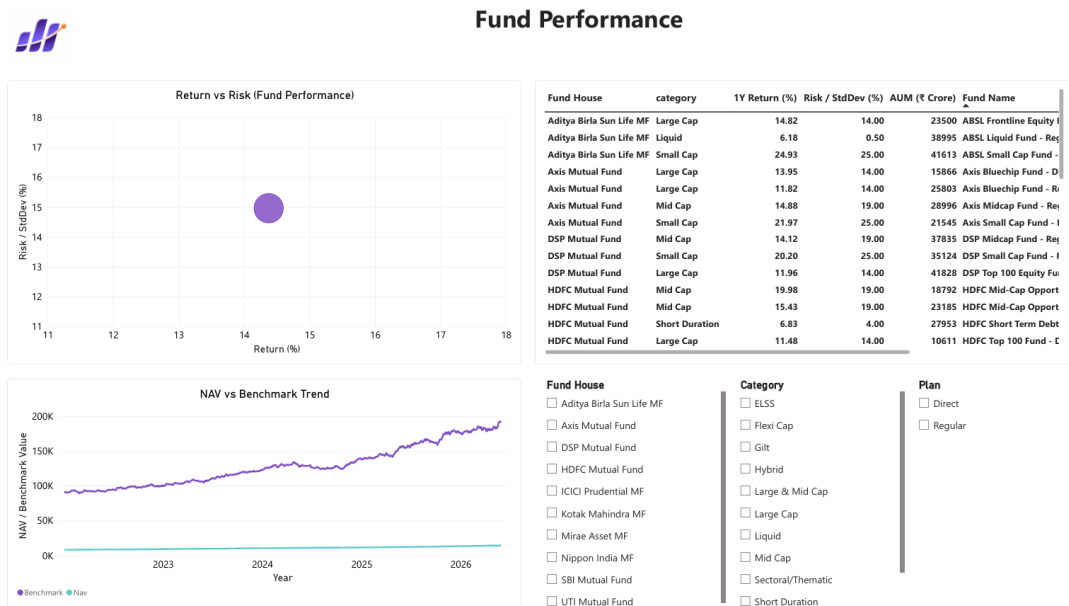


Figure 7.2: Power BI Dashboard – Fund Performance

7.4 Investor Analytics

The third page focuses on investor transactions, geographic distribution, transaction types, age groups and transaction volume.

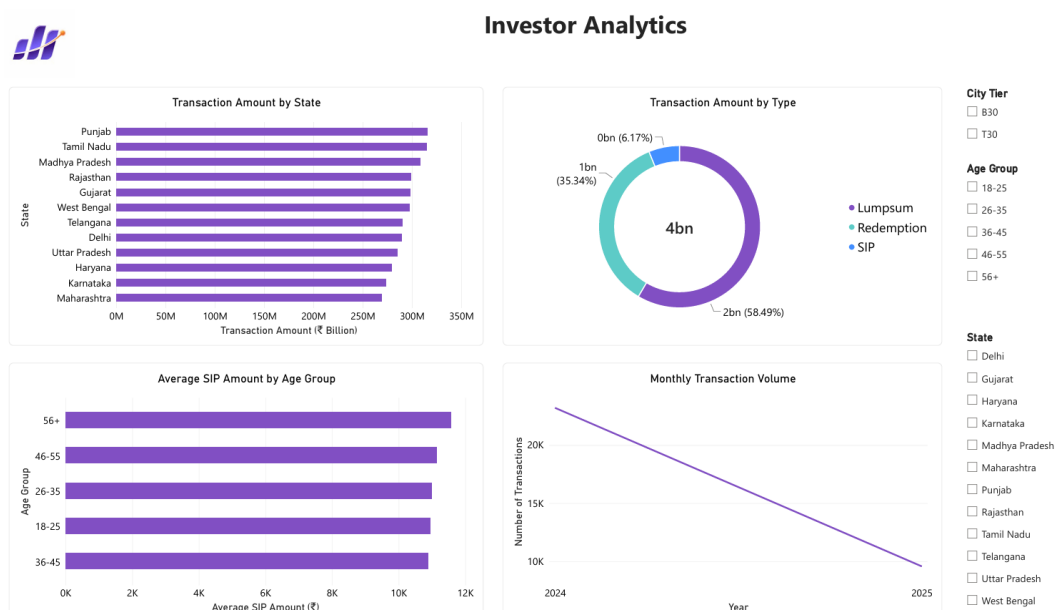


Figure 7.3: Power BI Dashboard – Investor Analytics

Transaction amounts are analysed by state, while a donut chart shows the distribution of SIP, Lumpsum, and Redemption transactions. Age-group analysis and monthly transaction volume provide additional insights into investor participation and transaction behaviour. Slicers allow filtering by state, age group, and city tier.

7.5 SIP and Market Trends

The fourth page combines SIP inflow trends with market benchmark information and category-level analysis. A dual-axis chart compares monthly SIP inflows with the NIFTY 50, while a category heatmap highlights monthly inflow patterns. A bar chart presents the top five categories by net inflow for FY25, and a KPI displays year-over-year SIP account growth.

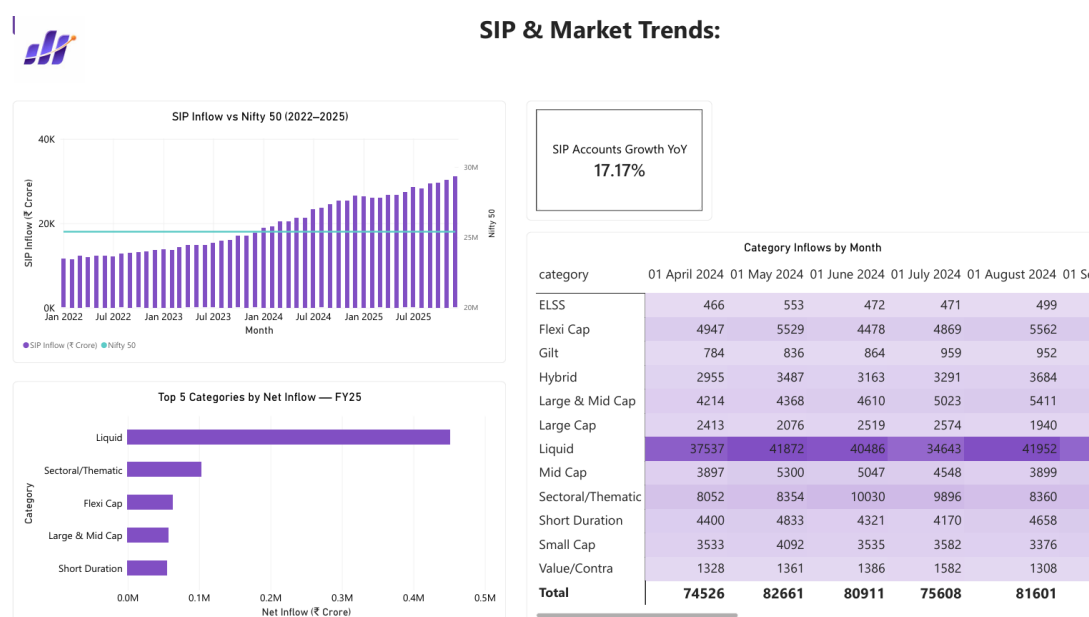


Figure 7.4: Power BI Dashboard – SIP and Market Trends

7.6 Interactivity and Branding

Interactive tooltips and drill-through functionality were added to improve dashboard navigation. Users can drill through from the fund scorecard to detailed NAV and benchmark information. The dashboard was formatted using Bluestock branding, including the Bluestock logo, consistent colours, titles and visual formatting across all four pages.

Chapter 8

Findings and Conclusions

8.1 Key Findings

- Mutual fund industry AUM showed substantial growth during the analysed period.
- SIP inflows demonstrated a strong long-term growth trend, reaching approximately Rs.31,002 Cr in December 2025.
- Mutual fund folios increased from 13.26 Cr in January 2022 to 26.12 Cr in December 2025.
- Fund schemes showed different levels of return, volatility and risk-adjusted performance.
- Sharpe, Sortino, Alpha, Beta, and Maximum Drawdown highlighted differences in fund quality and risk.
- Benchmark comparison provided additional context for evaluating fund performance.
- VaR and CVaR highlighted differences in downside risk across schemes.
- Investor behaviour varied across age groups, transaction types and geographic locations.
- Sector concentration varied across portfolios, indicating different levels of diversification.
- The composite fund scorecard helped compare schemes using multiple performance and risk factors.

8.2 Recommendations

- Evaluate mutual funds using both return and risk metrics rather than return alone.
- Use Sharpe, Sortino, Alpha, Beta, and Maximum Drawdown together for a comprehensive assessment.
- Monitor funds with relatively high downside risk using VaR and CVaR.
- Encourage consistent SIP investing for long-term wealth creation.
- Monitor investors showing longer gaps between SIP transactions.
- Review portfolios with high sector concentration to improve diversification.

8.3 Limitations

- Historical performance does not guarantee future returns.
- Analysis depends on the quality and completeness of the available datasets.
- The selected risk-free rate is a proxy assumption.
- Historical VaR may not fully capture future extreme market events.
- The recommendation model is simplified and should not be considered financial advice.
- Benchmark selection can influence performance comparison.
- Some industry-level values may differ from current real-time market values.

8.4 Conclusion

This project demonstrates an end-to-end mutual fund analytics workflow covering data ingestion, ETL, exploratory analysis, financial performance measurement, risk analysis, advanced analytics and business intelligence reporting.

The combination of Python, SQL, SQLite, financial analytics and Power BI provides a practical framework for transforming raw mutual fund data into meaningful insights for performance evaluation, investor analysis and risk assessment.